

LESSON 2-3 • SESSION 1 • PRACTICE • FORM B

Median and Outliers

Learning target – I can explain why the median is a better measure of center than the mean when a data set has an outlier.
• 6.SP.B.5d

REMEMBER

- An **outlier** is a value far from all the others. It can be very high or very low.
- An outlier **pulls the mean** toward it. The **median** stays in the middle of the group.
- With an outlier, the median is the better measure of a **typical** value.

WORKED EXAMPLE

Car counts: 44, 39, 41, 47, 43, 45, 120.

Step 1 Order: 39, 41, 43, 44, 45, 47, 120.

Step 2 Outlier: 120. Median: 44 (the 4th value).

Step 3 Mean: $379 \div 7 \approx 54$, greater than 6 of the 7 counts.

Answer Typical Saturday: about 44 cars (the median)

WORDS outlier (*valor atípico*) • median (*mediana*) • mean (*media*)

PRACTICE

show your work • label units

1. **Cup Sales.** Seven weekends brought 52, 48, 50, 55, 51, 53, 130 cups.

Hint Order first. The outlier is far from all the others.

- a. Outlier: _____ b. Median: _____ c. Typical value to report: _____

2. **Long Jumps.** Seven jumps went 41, 44, 43, 46, 42, 45, 85 inches.

a. Outlier: _____

b. Median: _____

3. **Missed Questions.** Seven students: 7, 9, 8, 10, 6, 5, 48.

Hint Does the middle value move?

a. Median: _____

b. Median if 48 becomes 10: _____

4. **Museum Tickets.** Seven tickets cost 26, 31, 24, 29, 33, 28, 140 dollars. Find the median price. Why does it describe a typical ticket better than the largest price?

Show your work.

Answer _____ dollars

5. **Bigger Outlier.** A set of 11 wait times has a median of 35 minutes. Its largest value, 240 minutes, changes to 400 minutes. What happens to the median?

Hint Does the order of the values change?

- ☐ It goes up. ☐ It stays the same. ☐ It goes down.

Why _____

explain the mistake, then fix it

-

choose the best answer

- find your answer · copy its letter into that problem's box · spell a math word*

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____